

Ashwin Padaki

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SUMMARY

Computer Science Ph.D. student specializing in efficient data structures for nearest neighbor search and other problems involving high dimensional data. Also interested in practical aspects of vector retrieval systems.

EDUCATION

University of Pennsylvania

Ph.D in Computer and Information Science

Philadelphia, PA

2024 – 2029 (expected)

Advisors: Erik Waingarten and Sanjeev Khanna.

Columbia University

B.A. in Computer Science, Mathematics

New York, NY

2020 – 2024

INDUSTRY EXPERIENCE

Research Scientist Intern

Pinecone

Jun 2026 – present

New York, NY

Quantitative Trader Intern

Optiver

Jun 2022 – Aug 2022

Chicago, IL

PUBLICATIONS

Unless otherwise specified (*), authors are ordered alphabetically by last name.

- Sanjeev Khanna, **Ashwin Padaki**, Erik Waingarten. [Learning Partition Trees for Nearest Neighbor Search](#).
(in submission)
- (*) Tian Zhang, **Ashwin Padaki**, Jiaming Liang, Zack Ives, Erik Waingarten. [Prune, Don't Rebuild: Efficiently Tuning \$\alpha\$ -Reachable Graphs for Nearest Neighbor Search](#).
(in submission)
- Sanjeev Khanna, **Ashwin Padaki**, Erik Waingarten. [Sparse Navigable Graphs for Nearest Neighbor Search: Algorithms and Hardness](#).
SODA 2026
- Sanjeev Khanna, **Ashwin Padaki**, Krish Singal, Erik Waingarten. [A Polynomial Space Lower Bound for Diameter Estimation in Dynamic Streams](#).
FOCS 2025
- Henry Fleischmann, Kyrylo Karlov, Karthik C. S., **Ashwin Padaki**, Styopa Zharkov. [Inapproximability of Maximum Diameter Clustering for Few Clusters](#).
SODA 2025
- Josh Alman, Yunfeng Guan, **Ashwin Padaki**. [Smaller Low-Depth Circuits for Kronecker Powers](#).
SODA 2023

TALKS

1. Learning Partition Trees for Nearest Neighbor Search

• NYU Algorithms Seminar

May 2026
2. Sparse Navigable Graphs for Nearest Neighbor Search

• SODA 2026 Conference

Jan 2026

• NYC Graduate Student Theory Seminar

Nov 2025

• DIMACS/Rutgers Theory Seminar

Nov 2025

• CMU Theory Seminar

Sep 2025
3. Inapproximability of Maximum Diameter Clustering for Few Clusters

• SODA 2025 Conference

Jan 2025

SERVICE

Head Teaching Assistant <i>Algorithms for Big Data (UPenn)</i>	Sep 2025 – May 2026 <i>Philadelphia, PA</i>
Seminar Organizer <i>UPenn Theory Seminar</i>	Sep 2025 – present <i>Philadelphia, PA</i>
Seminar Instructor <i>Columbia Undergraduate Learning Seminar in Theoretical Computer Science</i>	Sep 2023 – Dec 2023 <i>New York, NY</i>

ACHIEVEMENTS

National Science Foundation (NSF) Graduate Research Fellow	2024
Phi Beta Kappa Inductee	2024
Putnam Mathematical Competition, Top 500 Scorer	2022